Competitive Pricing Optimization for Rideshare Services

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Abstract

This project analyzes a large Boston ride-share dataset to understand and optimize competitive pricing strategies for Uber and Lyft. We first conduct exploratory analysis to quantify a 9.8% average Lyft-Uber price gap and systematically test potential drivers: ride length, geographic coverage, surge behavior, and ride-type composition. Distance and geography are nearly identical across platforms, while surge multipliers and the Lux-WAV mismatch explain most of the disparity. We then build linear regression models to predict base fares, showing that distance and ride type dominate, with Lyft's explicit surge multiplier greatly improving price predictability. Finally, we apply k-means clustering on distance, base_price, surge, and hour to uncover four economically meaningful segments: two short shared commuter pools around Haymarket, a premium high-surge Back Bay-North Station business corridor, and a medium-long, high-value Boston University-Financial District segment. Cluster-specific regressions validate these segments and support targeted recommendations: stable, volume-focused pricing for commuter routes; aggressive surge and premium supply for business corridors; and partnership- and upsell-driven strategies for university and long-distance trips.

I. Introduction and Statement of the Research Question

Ride-sharing platforms such as Uber and Lyft rely heavily on dynamic pricing to balance supply and demand, manage congestion, and remain competitive. Prices vary by location, time of day, ride type, and sometimes weather conditions. However, it is not obvious how these platforms should tune dynamic pricing rules across different contexts to simultaneously maximize revenue per ride and maintain a competitive edge.

In this project, we use the Uber and Lyft Dataset - Boston, MA from Kaggle, which contains several hundred thousand ride records with information on the platform (Uber or Lyft), ride type, pickup and dropoff locations, timestamps, prices, surge multipliers, distance, and weather conditions. This dataset provides a rich view of how trips and prices behave in a major urban market.

Our central research question is: "How should Uber and Lyft optimize dynamic pricing strategies across different geographic locations, times of day, and weather conditions to maximize revenue per ride while maintaining a competitive edge?"

We address this question in three main steps:

1. Data exploration. We examine the structure and quality of the data, summarize key categorical variables (platform, ride type, locations), study the distribution of prices and surge multipliers, and identify the main drivers of price (distance, ride type, time, weather). We also engineer a new "base_price" variable that separates underlying fare from surge.

2. Model building.
    a. First, we fit linear regression models (including regularized variants) to predict base price from distance, platform, ride type, time, and weather. This provides a "global pricing engine" and quantifies which features matter most.
    b. Second, we apply k-means clustering to key pricing features (distance, base price, surge multiplier, hour of day) to uncover market segments - groups of rides with similar pricing behavior - and then validate these clusters with cluster-specific regressions.

3. Interpretation and recommendations. We combine the regression and clustering results to develop segment-specific pricing recommendations for Uber and Lyft. In particular, we propose different strategies for high-volume commuter pools, premium business corridors, and university / financial-district segments.

The rest of the report is organized as follows. Section 2 describes the dataset, preprocessing, and exploratory data analysis (EDA). Section 3 presents the model building, including both the linear regression (pricing model) and the clustering (market segmentation). Section 4 summarizes our findings. Lastly, section 5 documents computational considerations.

II. Data Exploration

2.1 Dataset Description and "Bigness"

We use the Uber and Lyft dataset for Boston, MA from Kaggle (https://www.kaggle.com/datasets/brllrb/uber-and-lyft-dataset-boston-ma), which compiles ride records collected through APIs and logging tools. The full dataset contains thousands of trips (on the order of 500,000 in our training sample), each with:

- Platform (cab_type): Uber or Lyft.
- Ride type (name): 12 categories, including economy and shared options (e.g., Lyft, Lyft XL, Lyft Shared, UberX, UberPool, UberXL) and premium services (e.g., Black, Black SUV, Lux, Lux Black, Lux Black XL, WAV).
- Location: 12 pickup locations (source) and 12 destination locations (destination), representing zones in the Boston area (e.g., Back Bay, Financial District, South Station, North End, Haymarket Square, Boston University).
- Price and surge: price (the observed fare) and surge_multiplier.
- Trip characteristics: distance (approximate trip distance in miles).
- Temporal variables: timestamp from which we derive hour of day, day of week, and other time features.
- Weather variables: short_summary (e.g., Clear, Overcast, Light Rain), temperature, and other conditions.

The dataset is "big" in the sense that it contains a large number of observations and moderate dimensionality, making analyses computationally nontrivial. However, it is still small enough to fit into memory on our machines, so we can work with it directly after some filtering and careful use of vectorized operations and aggregation.

2.2 Preprocessing and Feature Engineering

We performed several preprocessing steps:

1. Handling missing and invalid values.
    a. We removed observations where the price of a ride is missing. This ensures that downstream models (both regression and clustering) are fit on complete and interpretable records.

2. Platform and ride-type factors.
    a. We treated cab_type (Uber vs Lyft) and name (ride type) as categorical variables (factors) for modeling. This allows us to estimate distinct effects for economy vs premium products for Uber/Lyft.

3. Base price variable.
    a. A crucial insight from our EDA is that the surge_multiplier behaves very differently by platform. In the Boston dataset, Lyft exhibits a wide range of surge multipliers (> 1), whereas Uber's surge_multiplier is always exactly 1. This means that the raw price mixes underlying fare with surge only for Lyft, but not for Uber.
    b. To make prices comparable across platforms, we define: base_price = price / surge_multiplier, for rides with non-missing and non-zero surge_multiplier. This base_price is our estimate of the non-surge, underlying fare. It becomes the main target variable in our modeling and clustering, allowing us to separate structural pricing (distance + ride type) from dynamic surge.

4. Time variables.
    a. We converted timestamps into interpretable fields such as hour of day and day of week. Hour (0-23) was used directly in clustering and regression as a simple proxy for time-of-day effects (e.g., early morning vs mid-day vs late night).

2.3 Categorical Structure: Platforms, Ride Types, and Locations

We first examined the categorical structure of the data:

- Platform distribution.
    - Rides are roughly split between Uber and Lyft (close to 50-50 in our training sample), providing enough data to compare the two platforms across ride types and locations.

- Ride-type distribution.
    - There are 12 distinct ride types, and in this sample they are relatively evenly distributed: each service (e.g., Lyft, Lyft XL, Lyft Shared, Lux, Lux Black, Lux Black XL for Lyft, and UberPool, UberX, WAV, UberXL, Black, Black SUV for Uber) has on the order of roughly 40,000 rides. This balanced distribution gives us enough data to compare pricing structures across all tiers, from shared and economy to premium services.

- Geographic structure.
    - Trips occur between 12 pickup locations and 12 destination locations. A handful of routes (e.g., South Station <-> Financial District, Back Bay <-> North Station, Boston University <-> Financial District, Haymarket Square loops) account for a large fraction of rides. These recurring routes later motivate our clustering by route characteristics and our route-specific pricing recommendations.

2.4 Price, Surge, and Base Fare

The distribution of price is right-skewed: most trips fall in a relatively low to moderate price range, but there is a long tail of expensive rides. This has two implications: we must be robust to outliers in modeling, and from a business perspective, a small fraction of rides generate a disproportionate share of revenue.

The surge multiplier exhibits a notable platform asymmetry:

- For Lyft, surge_multiplier takes values greater than 1 for many rides, indicating that dynamic surge pricing is directly recorded.
- For Uber, surge_multiplier is always 1 in this dataset, suggesting that either Uber's surge is not recorded the same way or that we are observing mostly non-surge periods for Uber.

Because of this, the raw price combines two different mechanisms for Lyft (base fare + surge) but effectively only base fare for Uber. The engineered variable base_price = price / surge_multiplier corrects for this and allows us to analyze underlying fares on a comparable scale.

Our Exploratory analysis shows that:

- base_price is strongly related to distance.
- base_price differs systematically across ride types, with shared/economy rides having low base prices and premium rides having high base prices.
- base_price varies less dramatically by hour or weather, although there are modest peaks around commute times and small shifts by weather condition.

These findings guide our model selection: distance and ride type are primary structural drivers; time and weather play secondary roles; surge is primarily a multiplicative factor on top of base_price rather than a core determinant of it.

2.5 Price Disparity Analysis

There is a significant price disparity between the average price of Lyft and Uber (9.8%). It is imperative to analyze the root of this difference to better understand the difference in pricing strategies and give meaningful suggestions.

We hypothesized that the price disparity could come from: different average ride lengths, different geographical focuses, different price surge strategies and different composition of ride types. By exploring the summary statistics, checking the visualization plots, and fitting some basic regressions, we were able to test our hypothesis.

2.5.1 Different Average Ride Length

We hypothesized that Lyft could have a longer average ride length compared to that of Uber, which drives up its price.

We test this hypothesis by finding the mean and the median distance of all rides, grouped by Lyft and Uber.

The result shows that the mean of the ride for Lyft and Uber is both 2.19 miles, and the median of the ride for Lyft and Uber is 2.14 and 2.17, respectively. With a relative difference of less than 2%, we concluded that the difference in distance is negligible, and it is not the primary reason behind the price disparity.

2.5.2 Different Geographical Focuses

We hypothesize that Lyft and Uber could have different level of presence for different areas in the city. Perhaps Lyft is gaining more market share over the more profitable routes.

We test this hypothesis by creating a 12-by-12 grid for Lyft and Uber, respectively, which shows the percentage of Lyft and Uber ride for a certain source/destination pair. We then calculated the relative difference for each 12x12 element from the two 12-by-12 grid.

We found that the greatest relative difference is only 1.6%, which shows that Lyft and Uber are comparable in terms of geographic coverage. Our conclusion is that there is no substantial difference in geographic focuses between Lyft and Uber, and it is not the primary source of the price disparity.

2.5.3 Different Price Surge Strategies

We hypothesize that the price disparity could come from the surge_price structure of Lyft, which would increase the price by a factor that results to an overall higher price.

We tested this hypothesis by removing the surge_price of Lyft rides and compare the base_price of the Lyft and the Uber rides.

The result shows that the previous price disparity of 9.8% were reduced to 6%, which suggests that the surge multiplier accounts for part of the higher price of Lyft rides, but not all.

2.5.4 Different Ride Compositions

We hypothesize that the price disparity could come from the difference in ride compositions: perhaps more high-end rides offered by Lyft compared to uber.

We tested this hypothesis first by checking the type of rides for both Lyft and Uber.

We noticed that in addition to the fact that both Uber and Lyft offers six different ride types, the amount of rides in each type are very close to other types of rides for a given company (Uber or Lyft). So both the observations for Uber and Lyft are evenly divided into six ride type categories.

It first seems that this even distribution suggests that this is not the driving force between the price disparity, but when we dive deeper into the comparable pairs between Uber and Lyft ride types, we noticed that although five out of six types of the two companies has a comparable pair (in terms of naming) against each other, namely:

- Shared (Lyft) vs. UberPool (Uber)
- Lyft vs. UberX
- Lyft XL vs. UberXL
- Lux/Lux Black vs. Black
- Lux Black XL vs. Black SUV

Aside from the technical detail that Black from Uber could compare to either Lux or Lux Black from Lyft, there is not a clear counterpart for the WAV from Uber.

To investigate this further, we fit twelve simple regression model, one for each ride type, of predicting the price using the distance. We call the intercept as the base fare, and the slope as the per-mile rate. We noticed that aside from the significant base fare difference between the carpool level of two companies and the WAV-Lux comparison, all the base fare and per-mile relative difference is roughly within 30%. But for the WAV-Lux comparison, the base fare is 39.1% more expensive for Lyft Lux, the per-mile rate is shockingly 161.9% more expensive for Lyft Lux, and the overall average price difference for this pair is 81.6%.

By removing this pair from the average price calculation, we found that the price difference reduced from the original 9.8% to 1.6%, which shows that most of the price disparity comes from this lopsided comparison pair.

2.6 Insights Carried into Modeling

The EDA provides several key insights that directly inform our subsequent modeling choices:

1. Choose base_price as the main response.
    a. Due to the asymmetric recording of surge across platforms, modeling price directly would conflate underlying fare with surge only for Lyft. Using base_price as the response allows a fair comparison and isolates structural fare differences.

2. Use distance and ride type as core predictors.
    a. Distance and ride type explain much of the variation in base_price, so any pricing model should include them prominently.

3. Treat platform and time as modifiers.
    a. Platform (Uber vs Lyft) and hour of day have detectable but secondary effects, suggesting interactions (e.g., platform-by-ride-type or platform-by-distance) may be useful in regression.

4. Motivate clustering on pricing-related features.
    a. The repeated routes, clear product tiers, and differences in base_price and surge across contexts motivate our use of clustering later on: we expect that rides will naturally group into distinct pricing segments (e.g., commuter routes vs premium business routes), and these segments may suggest different pricing strategies.

III. Model Building

3.1 Linear Regression

3.1.1 Motivation for Regression Analysis

With the findings in EDA, we want to know more about their pricing strategies in order to find rooms of improvement. We decided to take a deeper dive into the pricing structure of Uber and Lyft to extract more information about the difference in their pricing strategy. To achieve this, we fitted several linear regression models to predict the price of a ride using various features.

3.1.2 Baseline Model

We choose to use the following features to create a baseline regression model to predict the price of a ride:

- Distance: the distance of a ride
- Surge Multiplier: a factor that is multiplied into the price
- Cab Type: twelve cab types
- Temperature: an aggregated temperature value
- Wind Speed: an aggregated wind speed value
- Weather: a categorical variable for the type of weather
- Time: day of the week * hour of the day

This eventually yielded an R^2 of 0.928, which suggests that the basic model is predicting at a reasonably well level when giving some of the features.

3.1.3 Cab-Type-Stratified Model

We incorporated our findings of the price disparity among cab types, especially for the per-mile rate part, into this model by using the interaction term of distance * cab type.

Using this model, the R^2 increased from 0.928 to 0.951, which suggests that it is imperative to recognize the difference in per-mile rate and incorporate it into our model.

3.1.4 Log-Transformed Model

In addition, we used a log transformation for the price, distance and the surge multiplier to better capture the multiplicative nature of the price calculation and the skewed distribution of the price.

The resulting R^2 increase from 0.928 to 0.939, suggesting that using log-transformation has some help in handling the skewed distribution and the multiplicative nature of the pricing calculation

3.1.5 Combining Interaction and Log-Transformation

We tried to combine the previous two improvement ideas and fit a more sophisticated model. But we noticed that the R^2=0.945 for this model, which is between the R^2 for the two previous models. We think perhaps the log transformation is disrupting the interaction term for capturing a clear per-mile rate difference. We eventually decided to skip this model.

3.1.6 Lasso Feature Selection Model

To further streamline the model, we performed the Lasso selection on our previous model of using the interaction term. The result shows that all the distance and cab type features, surge_multiplier, and a subset (about 16%) of the temporal features is important. This suggests that:

- The distance and cab type is main driver of predicting the price.
- Surge multiplier is an important tool in breaking down the base price to the surge in prices for Lyft
- Temporal data has some importance, but is less crucial compared to the distance and potential price surge events.
- Weather data is not significant, perhaps due to the nature of the dataset: limited three-week timeframe in a winter.

For the R^2 of this model, it is the same with the non-reduced model, which is 0.951, suggesting that the model selection are pruning out non-contributing features successfully.

3.1.7 Other Models

We tried to add more information into the model, such as a further breakdown of the hour to 15-minute intervals, and incorporate source-destination pair information, but the results of such modifications have little improvement compared to the Cab-Type_Stratified Model. We decided to follow our findings and take the suggestions from Lasso selection to limit the model complexity as is.

3.1.8 Further Analysis on the Surge Multiplier

One key finding about the surge multiplier is that only lyft has it. We then decide to check what will tell us more about the pricing strategy between the two.

To begin with, we fitted two models of predicting the price using distance for each cab type, such that one included the surge_multiplier, another is not. We then proceeded to check how well the first one outperformed the second one. To our surprise, by adding the surge multiplier, the R^2 for Lyft cab types increased dramatically, especially for the most luxurious ride: from 0.486 to 0.820. And the mean R^2

increased from 0.624 to 0.852 (excluding shared), while the R^2 of Uber remained as 0.636. This suggests that without the surge multiplier information, the price predictability between Uber and Lyft is pretty similar, and at a moderate level, but with the information, we can predict the Lyft dramatically well.

Essentially this is telling us Lyft adopts a more transparent surge pricing strategy whereas Uber is hidden in other factors. This disparity is a great starting point to look into the surge pricing strategy and potentially give recommendations to both platforms.

Detailed R^2 chart:

Service	Ride Type	R^2 (Raw)	R^2 (Base)	Improvement
Lyft	Lux	0.7011	0.8863	0.1852
Lyft	Lux Black	0.6615	0.8926	0.2311
Lyft	Lux Black XL	0.4856	0.8200	0.3344
Lyft	Lyft	0.6071	0.7969	0.1898
Lyft	Lyft XL	0.6626	0.8620	0.1993
Lyft	Shared	0.4123	0.4123	0.0000
Uber	Black	0.8339	0.8339	0.0000
Uber	Black SUV	0.7403	0.7403	0.0000
Uber	UberPool	0.4494	0.4494	0.0000
Uber	UberX	0.5404	0.5404	0.0000
Uber	UberXL	0.6171	0.6171	0.0000
Uber	WAV	0.5396	0.5396	0.0000

To continue, we tried to predict under what circumstance will Lyft issue surge_multiplier. Unfortunately, we tried multiple ways of regression and clustering, but failed to find a reasonably well prediction model.

Our exploration idea is to find the demand spikes and hypothesize that surge_multiplier will arise when there is a demand spike. We define demand spikes as a period of time in a certain location, there are significantly more rides starting from this place compared to the same time at another week or before/after this period of time. We use a period of 15 minutes as interval, and stratified based on all 12 source locations. Our model did find several demand spikes, but when we check the percentage of surge_multipliers with the general rate, it is pretty much pure guessing. We also tried to tune the parameters and tried some clustering ideas, but it is hard to predict surge_multipliers with a high level confidence.

Our hypothesis for why it is hard to predict is we only have successful rideshare data in this dataset. We don’t have the data of how many customers asked for a ride in a certain area at a certain time, nor do we have the data of how many drivers are in an area at a certain time. Instead a spike in demand, we should focus more on finding a spike in the gap between demand and supply. But without the information on the supply side, it is hard to predict surges using only the demand side.

3.2 Clustering for Market Segmentation

3.2.1 Motivation for Clustering

The regression model (developed in the previous section) answers a key question: Given the features of a trip (distance, platform, ride type, time, etc.), what is a reasonable base price? However, it does not directly address market segmentation: which sets of trips form coherent groups that might justify different pricing and operational strategies?

Dynamic pricing decisions are often made at the level of segments, not individual trips. For example, platforms may want separate rules for:

- High-volume commuter pools,
- Premium business corridors,
- University or event-driven travel patterns.

To uncover such segments, we apply k-means clustering to relevant features and then interpret the resulting clusters in terms of routes, ride types, and time of day. Clustering provides an unsupervised view of the data that complements the supervised regression: regression quantifies how features affect price; clustering groups trips with similar joint patterns of distance, price, surge, and time.

3.2.2 Intuition for K-Means

We chose k-means for several reasons. First, it is computationally efficient for large numeric datasets. Secondly, our main clustering features (distance, base_price, surge_multiplier, hour) are numeric. Third, it provides interpretable clusters via cluster centroids and within-cluster averages.

Guided by our EDA and regression results, we selected the following features for k-means:

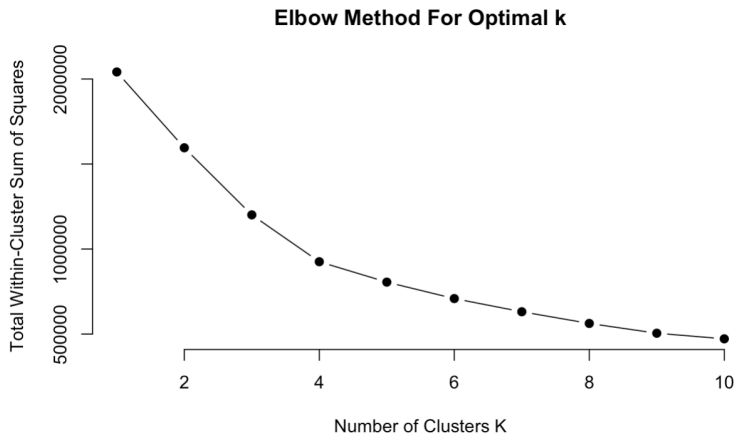
- distance,
- base_price (price divided by surge_multiplier),
- surge_multiplier,
- hour (0-23).

These features jointly capture key pricing dimensions: how far a trip goes, what its underlying fare is, how surge is applied, and when during the day it occurs.

We filtered out records with missing or non-finite values in these variables and then standardized each feature (mean 0, variance 1) before clustering.

3.2.3 Choosing the Number of Clusters

We ran k-means for k=1,2,3, …, 10 and computed the total within-cluster sum of squares for each k. By looking at the elbow plot (below), the within-cluster sum of squares decreases rapidly as k increases from 1 to 4, and the ains taper off after k=4. This suggests that 4 clusters strikes a useful balance between compactness and simplicity of interpretation.



We therefore fit a final k-means model with k=4 on the standardized features, using multiple random starts to reduce the risk of local minima.

3.2.4 Cluster Characteristics and Interpretations

Clusters 1 and 4 are very similar in their core statistics and thus are best interpreted together:

Cluster 1:

- Size: approximately 186,000 rides (one of the largest clusters).
- Average distance: about 1.8 miles.
- Average base_price: around $12.9.
- Average surge_multiplier: essentially 1.0.
- Average hour: early morning (around 5 a.m.).
- Most common source/destination: Haymarket Square.
- Most common ride type: UberPool.

Cluster 4:

- Size: approximately 194,000 rides (also very large).
- Average distance: about 1.8 miles.
- Average base_price: about $12.8.
- Average surge_multiplier: near 1.0.
- Average hour: early evening (around 5-6 p.m.).
- Most common source/destination: again, Haymarket Square.
- Most common ride type: UberPool.

These clusters represent short, shared commuter trips around Haymarket Square during the morning and evening rush hours. They are high-volume, low-surge, low-to-moderate fare rides dominated by pooled services. From a business perspective, these trips contribute a large share of total rides but relatively low revenue per ride.

Cluster 3 has the following characteristics:

- Size: approximately 122,000 rides.
- Average distance: about 3.4 miles.
- Average base_price: around $26.7.
- Average surge_multiplier: close to 1.0.
- Average hour: late morning to mid-day (around 12 p.m.).
- Most common route: Boston University -> Financial District.
- Common ride types: premium services, notably Black SUV.

This cluster captures medium-long, high-value trips that are often associated with university and central business district travel. The longer distances and higher base prices indicate intrinsically higher fares; the dominant premium ride types reinforce the idea that these trips often involve higher willingness to pay, such as student groups, professionals, or airport and cross-town travelers. Unlike the commuter clusters, surge is not the main driver here; instead, distance and ride type are the primary determinants of base_price.

Cluster 2 is distinct in both size and pricing characteristics:

- Size: approximately 7,900 rides (the smallest cluster).
- Average distance: about 2.4 miles.
- Average base_price: around $19.2.
- Average surge_multiplier: about 1.69 (the highest among all clusters).
- Average hour: late morning (around 11-12).
- Most common route: Back Bay -> North Station.
- Most common ride type: Lux Black and other premium services.

This cluster represents a premium, high-surge business route. Trips are moderately long but carry high underlying fares and consistently elevated surge multipliers. The route (Back Bay to North Station) and the predominance of Lux Black suggest high-income, business-oriented travel, such as commuters to the financial district, travelers connecting to regional rail, or event-based trips. Unlike the commuter clusters, both base_price and surge are significant contributors to total revenue per ride in this segment.

3.2.5 Cluster-Specific Regression for Validation

To validate that these clusters correspond to genuinely distinct pricing segments, we conducted separate linear regressions within each cluster, using:
base_price = beta_0 + beta_1*distance + beta_2*surge_multiplier + beta_3*hour + beta_4*name for cluster 1, 2, 3, and 4. Here, name is a categorical indicator for ride type. This allows us to compare how distance, surge, and ride type affect base fares inside each segment.

Across all clusters, these models achieved high R^2 values (roughly 0.87-0.95), confirming that a linear structure continues to describe base_price well even after conditioning on cluster membership. The estimated coefficients display systematic differences across clusters that align with our qualitative interpretations:

- In the commuter clusters (1 and 4), the distance coefficients are positive and moderate (around $2 per extra mile), while the incremental effect of surge_multiplier on base_price is small. Ride-type

effects show substantial discounts for shared and economy services relative to the baseline ride type. This supports our view of these clusters as distance-driven, low-surge commuter segments dominated by pooled and economy rides.

- In the premium business cluster (2), the distance coefficient is larger (around $3.8 per mile), and premium ride types (e.g., Lux Black) carry large positive coefficients relative to baseline, reflecting substantial price premia. Surge_multiplier also plays a more pronounced role, consistent with the observed high average surge. These patterns support the interpretation of Cluster 2 as a high-income, premium corridor where both distance and product choice strongly affect base fares and surge is used aggressively.
- In the medium-long, high-value cluster (3), the distance coefficient remains large (around $3.3 per mile), and premium ride types such as Black SUV and Lux Black exhibit strong positive effects. Surge_multiplier has a smaller but non-negligible impact. This aligns with our view of Cluster 3 as a structurally high-fare segment driven primarily by trip length and premium product tier rather than frequent surging.
- Overall, the cluster-wise regressions provide quantitative support for our cluster interpretations: the 4 k-means clusters correspond to economically meaningful segments with different pricing structures.
- See Appendix Figure 1, 2, 3, and 4 for more details (coefficient estimates, statistics, and cluster assignment) of the validation regression models

IV. Conclusion

Our analysis shows that Uber and Lyft ues fundamentally different pricing strategies and that a "one-size-fits-all" approach to dynamic pricing is inefficient. To maximize revenue, both platforms need to account for the differences in their pricing models and the specific behaviors of different market segments.

Based on our regression models, we found that Lyft uses a more transparent surging strategy driven by explicit surge multipliers and lower base fares. Uber, in contrast, hides its surge pricing but offers a more competitive per-mile rate, especially for premium ride types. We suggest that Lyft should advertise its predictable pricing and focus on retaining luxury customers, while simultaneously expanding its budget market share. Uber, on the other hand, should market the affordability of its high-end options for longer trips and use its flexible pricing to attract hesitant customers during non-peak times.

Furthermore, our clustering analysis shows that different routes require different operational strategies. We identified three specific segments that drive the market:

- Commuter Routes (e.g., Haymarket Square): These are high-volume, short, and shared trips where riders are very price-sensitive. For these segments, we recommend focusing on reliability rather than high margins. Platforms could offer commuter passes, loyalty rewards, and keep base fares stable to maintain volume.
- Premium Business Corridors (e.g., Back Bay to North Station): This segment consists of high-income riders willing to pay for premium services. Here, platforms can justify aggressive

surge pricing. We recommend targeting driver incentives to these areas to ensure supply matches the high demand.

- Institutional and Long-Distance Trips (e.g., Boston University to Financial District)**:** Revenue here is driven by distance and car type rather than surge. Strategies for this segment should focus on upselling premium vehicle features and creating partnerships (such as student or corporate rates) rather than relying on dynamic pricing spikes.

In summary, while distance and ride type are the main predictors of price, the key to optimization lies in specificity. By moving from a general pricing policy to one that targets these specific geographic and temporal segments, Uber and Lyft can better align their supply with demand and maximize revenue per ride.

To reflect on our approach, one key strength is that simple, interpretable models already explain most of the variation in prices, and the same core drivers remain important within each cluster. However, our models have limitations. They are trained on a short winter window in a single city, and they observe only successful rides (no demand or supply counts). Clustering is also restricted to four numeric features, so it likely misses behaviorally important dimensions such as user type or driver availability.

Future work could improve on these in several ways. First, incorporating explicit demand and supply signals - such as request counts, driver log-ins, or queue lengths - would allow direct modeling of when and where surge is triggered. Second, extending the analysis to multiple cities and seasons would test how portable the estimated pricing structures and clusters are, and whether segment-specific strategies generalize. Finally, integrating user-level outcomes such as wait times, cancellations, or conversion rates would allow platforms to optimize not only revenue per ride but also service quality and long-run retention.

V. Efficient Computing

One of our computation bottleneck is the time to fit the regression models. To tackle this, we implemented two efficient computing methods: using a small sample dataset and using multiple cores.

We generated a sample dataset of 10,000 observations from the cleaned dataset, which is about 2% of the cleaned dataset. Before running anything on the main dataset, we would first run it on the sample dataset, which is quick to load and get the result. So we can check the preliminary results and test our code before running the lengthy regression on the full dataset.

We also utilized multiple cores to reduce the runtime of running regressions. In the regression part of our report, we have five major regression models that use different regression equations. We choose to run these five regression models in parallel so that the runtime is reduced from the sum of runtime for the five models into roughly the slowest runtime for the five models.

VI. Appendix

Figure 1:

	term	estimate	std.error	statistic	p.value	cluster
1	(Intercept)	15.630427715	0.121329863	128.825891	0.000000e+00	1
2	distance	1.902500906	0.004545936	418.505894	0.000000e+00	1
3	surge_multiplier	-0.686920861	0.120304020	-5.709875	1.132306e-08	1
4	hour	-0.022763731	0.001047025	-21.741339	1.126651e-104	1
5	nameBlack SUV	9.652338947	0.021476117	449.445253	0.000000e+00	1
6	nameLux	-2.887604901	0.019008205	-151.913602	0.000000e+00	1
7	nameLux Black	1.194303670	0.020038836	59.599453	0.000000e+00	1
8	nameLux Black XL	9.864396149	0.022495540	438.504537	0.000000e+00	1
9	nameLyft	-9.651480415	0.018129111	-532.374732	0.000000e+00	1
10	nameLyft XL	-4.752006674	0.018517792	-256.618425	0.000000e+00	1
11	nameShared	-12.909234280	0.017805332	-725.020704	0.000000e+00	1
12	nameUberPool	-10.132569927	0.017619858	-575.065356	0.000000e+00	1
13	nameUberX	-9.220234852	0.017612662	-523.500366	0.000000e+00	1
14	nameUberXL	-4.047070213	0.018063109	-224.051697	0.000000e+00	1
15	nameWAV	-9.225496335	0.017635741	-523.113628	0.000000e+00	1

Figure 2:

16	(Intercept)	8.924769090	0.191290600	46.655555	0.000000e+00	2
17	distance	3.780383484	0.022724277	166.358801	0.000000e+00	2
18	surge_multiplier	-0.352373010	0.101649082	-3.466564	5.299576e-04	2
19	hour	-0.007684745	0.003366655	-2.282605	2.248005e-02	2
20	nameLux Black	5.456317413	0.074300089	73.436217	0.000000e+00	2
21	nameLux Black XL	14.455172540	0.074476948	194.089217	0.000000e+00	2
22	nameLyft	-8.396658235	0.074540569	-112.645481	0.000000e+00	2
23	nameLyft XL	-2.429625254	0.074623478	-32.558456	2.492168e-218	2

Figure 3:

24	(Intercept)	14.750066516	0.219747698	67.122735	0.000000e+00	3
25	distance	3.359002217	0.009385482	357.893425	0.000000e+00	3
26	surge_multiplier	-0.466122076	0.215563539	-2.162342	3.059377e-02	3
27	hour	-0.002555620	0.001292335	-1.977521	4.798501e-02	3
28	nameBlack SUV	8.772984882	0.029288369	299.538186	0.000000e+00	3
29	nameLux	-3.455436097	0.035945674	-96.129400	0.000000e+00	3
30	nameLux Black	2.089526125	0.032207584	64.876835	0.000000e+00	3
31	nameLux Black XL	10.072147474	0.029963479	336.147460	0.000000e+00	3
32	nameLyft	-16.016358985	0.059358823	-269.822717	0.000000e+00	3
33	nameLyft XL	-6.701245132	0.041173761	-162.755234	0.000000e+00	3
34	nameShared	-20.283755740	0.074383293	-272.692359	0.000000e+00	3
35	nameUberPool	-18.552136778	0.060340056	-307.459722	0.000000e+00	3
36	nameUberX	-15.843424771	0.055146522	-287.296901	0.000000e+00	3
37	nameUberXL	-5.274553187	0.038715157	-136.240006	0.000000e+00	3
38	nameWAV	-15.798097301	0.055274074	-285.813874	0.000000e+00	3

Figure 4:

39	(Intercept)	15.095067151	0.120583123	125.183912	0.000000e+00	4
40	distance	1.879808392	0.004493644	418.326043	0.000000e+00	4
41	surge_multiplier	-0.695343427	0.118446092	-5.870548	4.350605e-09	4
42	hour	0.021925978	0.001035916	21.165785	2.558430e-99	4
43	nameBlack SUV	9.685061470	0.021073180	459.591833	0.000000e+00	4
44	nameLux	-2.924030951	0.018614754	-157.081360	0.000000e+00	4
45	nameLux Black	1.192658946	0.019666562	60.643998	0.000000e+00	4
46	nameLux Black XL	9.897174236	0.022174868	446.323924	0.000000e+00	4
47	nameLyft	-9.578466065	0.017664119	-542.255516	0.000000e+00	4
48	nameLyft XL	-4.695121279	0.018122045	-259.083409	0.000000e+00	4
49	nameShared	-12.808502915	0.017443918	-734.267554	0.000000e+00	4
50	nameUberPool	-10.031797405	0.017217897	-582.637791	0.000000e+00	4
51	nameUberX	-9.122569615	0.017298736	-527.354699	0.000000e+00	4
52	nameUberXL	-3.984631586	0.017718785	-224.881763	0.000000e+00	4
53	nameWAV	-9.105571362	0.017276575	-527.047254	0.000000e+00	4